

# Chitresh Yadav

Backend & Applied ML Engineer

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## SUMMARY

Software engineer focused on backend services and applied ML systems. Built a streaming speech-to-speech pipeline (ASR → MT → TTS over WebRTC) with measured 2505 ms mean / 2580 ms p50 end-to-end latency, then cut latency 23.6% with Silero-VAD segmentation. Fine-tuned Whisper with LoRA for Hindi-English code-mixed ASR and designed an ONNX export path for Rust inference. Also shipped queue-based media workers (Redis, PostgreSQL, MinIO, Docker Compose) and a token-efficient LLM router with OpenTelemetry. Seeking SDE-1 / Backend / Applied ML roles.

## SKILLS

**Languages:** Python, Rust, TypeScript / Node.js, SQL, Dart

**Backend:** FastAPI, Express.js, REST APIs, WebRTC (aiortc), asyncio, Redis queues, PostgreSQL, SQLite, MongoDB, MinIO / S3

**ML / AI:** Whisper / faster-whisper, LoRA / PEFT, ONNX Runtime (fp16 / int8), IndicTrans2, Silero-VAD, BART, LLM routing, WER / CER eval, OpenTelemetry

**Infra:** Docker, Docker Compose, GitHub Actions, GCP (Cloud Run, Cloud Storage, Firebase), Linux

## PROJECTS

**speech-relay — Real-Time Speech-to-Speech Translation** [github.com/csy20/speech-relay](https://github.com/csy20/speech-relay)

*Python · faster-whisper · IndicTrans2 · Coqui / edge-tts · aiortc · ONNX Runtime · asyncio*

- Built a streaming ASR → MT → TTS service that returns live translated audio over WebRTC (EN→HI), instrumented per hop at 2504.73 ms mean / 2579.7 ms p50 end-to-end.
- Replaced fixed-window flushing with Silero-VAD segmentation and tuned buffers, cutting end-to-end latency by 23.61%.
- Designed a fan-out path so one shared ASR stream serves N listeners with independent translation targets (1.56× parallel efficiency at 2 listeners).
- Quantized ASR/TTS models with ONNX Runtime fp16/int8 (up to 11.25% faster inference) and added a WER/CER plus latency evaluation harness.

**speech-relay-rust — Code-Mixed ASR Fine-Tune + ONNX Serving** [github.com/csy20/speech-relay-rust](https://github.com/csy20/speech-relay-rust)

*Python · Whisper · LoRA / PEFT · PyTorch · ONNX · Rust (ort) · MUCS / OpenSLR-104*

- Implemented a data pipeline over MUCS Hindi-English speech: 16 kHz resample, code-mixed script filter (Devanagari + Latin), train/val/test split, Hugging Face export.
- Fine-tuned Whisper with LoRA on decoder attention (q/k/v/out proj) so the model handles mid-utterance Hindi-English switches; training path runs on a T4 in a few hours.
- Designed the serving path: merge LoRA weights, export encoder/decoder to ONNX, int8 quantize, run inference from a Rust ONNX Runtime (ort) service outside Python.

**MediaPipe AI — Distributed Media Processing Pipeline** [github.com/csy20/mediapipe-ai](https://github.com/csy20/mediapipe-ai)

*Express · Python workers · Redis · PostgreSQL · MinIO · Docker Compose · Whisper · BART*

- Architected a 5-container system: React client → Express REST gateway → Redis job queue → Python workers (Whisper transcription + BART summarization) → PostgreSQL metadata + MinIO object storage.
- Added async job status polling, structured logging, and S3-compatible uploads for audio/video up to 500 MB so workers scale independently of the API.

**router-agent — Token-Efficient LLM Router** [github.com/csy20/router-agent](https://github.com/csy20/router-agent)

*Python · Docker · Ollama · Fireworks API · OpenTelemetry / SigNoz · AMD Hackathon ACT II*

- Built a hybrid router for the AMD Developer Hackathon that classifies tasks (math, NER, summarization, code, QA) with zero-token heuristics, then escalates to a local 1.5B LLM or at most one paid API call.
- Instrumented classify → local → API stages with OpenTelemetry traces into SigNoz and shipped the agent as a Docker service with schema validation.

**ByteWise — Learning App (shipping proof)** [play.google.com/store/apps/details?id=com.csy20.bytwisewise](https://play.google.com/store/apps/details?id=com.csy20.bytwisewise)

*Flutter · Dart · Riverpod · Drift / SQLite · Google Play*

- Shipped a production Flutter app to Google Play with 788+ DSA / system-design lessons and an offline-first in-memory + SQLite cache. [Play Store listing](#).

## EDUCATION

**B.Tech, Computer Science and Business Systems**

Shri Shankaracharya Technical Campus, Bhilai, Chhattisgarh

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